Common Spatial Pattern Variants for Feature Extraction in Multi-Class Motor Imagery BCI

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Abstract*— Brain-Computer Interfaces (BCIs) based on Motor Imagery (MI) offer intuitive control but face challenges due to noisy electroencephalography (EEG) signals that often contain artifacts, necessitating effective feature extraction. Common Spatial Pattern (CSP) is a standard technique, yet its limitations in multi-class MI scenarios motivate various extensions. This paper investigates the classification performance, computational cost, and the crucial performance versus cost trade-off inherent in practical BCI applications of five prominent CSP variants, including standard CSP, Regularized CSP (RCSP), Filter Bank CSP (FBCSP), Sparse CSP, and Kernel CSP, when applied to multi-class MI classification. Using the PhysioNet MI dataset, we employed a consistent preprocessing pipeline (18 sensorimotor channels, filtering, epoching) and evaluated each variant by feeding extracted features into a Support Vector Machine (SVM) classifier with an RBF kernel. The evaluation revealed that standard CSP and RCSP achieved the highest peak average accuracy (~97.8%) with near-efficient computational cost. FBCSP yielded high accuracy (~96.7%) but incurred the highest computational cost. Sparse CSP was computationally efficient but achieved lower peak accuracy (~95.3%), while Kernel CSP showed the lowest peak accuracy (~92.8%) at moderate cost. The findings highlight the performance profiles and efficiency of these CSP methods for this multi-class MI task, indicating that standard CSP and RCSP offer a particularly effective combination of high accuracy and manageable computational load under the evaluated conditions.

***Index Terms*— Brain-Computer Interface (BCI), Motor Imagery (MI), Electroencephalography (EEG), Common Spatial Pattern (CSP), Feature Extraction, Multi-class Classification, Support Vector Machine (SVM).**

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) detect, process, and decode brain activity into commands, enabling applications in gaming, entertainment, security, education, and marketing [1]. However, the medical field remains the primary focus, with applications in communication, rehabilitation, diagnosis, and assistive technologies [2]. Motor Imagery (MI), where users control BCIs by imagining movement, is a common and intuitive paradigm [3].

MI-BCIs typically use electroencephalography (EEG) to detect sensorimotor rhythm changes. EEG is advantageous due to being non-invasive, affordable, and portable [4]. Yet, developing robust MI-BCIs is challenging because EEG signals are noisy, high-dimensional, often contain artifacts, and exhibit high inter or intra-subject variability [5]. Effective feature extraction is thus crucial to identifying user intentions from complex raw signals [1].

The Common Spatial Patterns (CSP) algorithm is a cornerstone feature extraction method for MI-BCIs [6]. This supervised technique finds spatial filters maximizing variance differences between MI classes, enhancing separability. Its popularity stems from high accuracy and low computational cost, demonstrated in numerous studies [7], [8].

While effective for two-class tasks, standard CSP performance degrades in multi-class scenarios (more than two imagined movements), which are vital for versatile BCI control. Limitations include handling data complexity, potential overfitting, noise sensitivity, and its binary-focused design, motivating CSP variants [9].

To address these shortcomings, several variants of the standard CSP algorithm have been proposed. Regularized CSP (RCSP), for instance, introduces regularization to improve robustness against noise and limited training data. The effectiveness of RCSP in combination with channel selection methods has been demonstrated [10]. Further work showed that correlation-based channel selection combined with RCSP could significantly improve classification accuracy [11]. Filter Bank CSP (FBCSP) extends CSP by applying it across multiple frequency bands to capture a wider range of discriminative information. FBCSP was utilized with adaptive time windows [12], achieving significant improvements in accuracy compared to existing methods. The FBRTS method, combining filter banks with Riemannian tangent space features, was proposed [13], outperforming FBCSP in multi-category MI-BCIs. Kernel CSP, on the other hand, utilizes kernel methods to map the data into a higher-dimensional space, potentially capturing non-linear relationships. Multi-class Kernel CSP (MKCSP) was introduced [9], while a Kernel CSP method using spatial covariance to reduce computational complexity was developed by [14]. Furthermore, a sparse CSP method using recursive weight While each variant offers potential improvements, they also present their own challenges, such as increased computational complexity or sensitivity to parameter selection.

Crucially, comprehensive, standardized comparisons of these variants under identical conditions are lacking. Existing studies use varied datasets and methods, hindering direct comparisons and informed decisions about the best variant for specific multi-class MI-BCI applications.

Therefore, this paper investigates the performance and computational demands of five CSP variants (CSP, RCSP, FBCSP, Kernel, Sparse) for multi-class MI classification. Using a standardized methodology (dataset, preprocessing, SVM classifier, evaluation metrics), we assess classification accuracy, computational cost, and the performance-cost trade-off. These results aim to provide benchmarks, guide algorithm selection, and inform future MI feature extraction research.

II. Methods

A. Data

For this study, we utilized the publicly available EEG Motor Movement/Imagery Dataset from PhysioNet [16], [17], which contains 64-channel EEG recordings from 109 subjects. Due to computational cost, we focused our analysis on data from 10 randomly selected subjects performing cued motor imagery tasks corresponding to three distinct classes, mapped to labels T0, T1, and T2, representing tasks including opening and closing left or right fist and both fists or both feet, as defined in the dataset documentation. Data were recorded at a sampling rate of 160 Hz using the Enhanced European Data Format (EDF+). From each selected subject, we used only the six experimental runs corresponding to motor imagery (runs R04, R06, R08, R10, R12, and R14).

B. Preprocessing

The raw EEG data underwent several preprocessing steps using the MNE-Python library [18] to reduce the noise and isolate brain signals related to the motor imagery tasks. Initially, each EDF+ file was loaded, and channel names were standardized. Only 18 channels located over the sensorimotor cortex and considered relevant for motor imagery were retained, including FC5, FC3, FC1, FC2, FC4, FC6, C5, C3, C1, C2, C4, C6, CP5, CP3, CP1, CP2, CP4, and CP6 [19]. A standard 10-10 montage was then applied. The data were band-pass filtered between 8 Hz and 30 Hz to maintain frequencies relevant to brain activity and a notch filter at 60 Hz to remove powerline interference. The filtered data were downsampled to 100 Hz to reduce computational load and then re-referenced using the common average reference (CAR) method across the selected channels. Finally, the continuous data were then segmented into epochs time-locked to event markers derived from the annotations embedded within the raw data files. Each epoch started at the event onset (tmin=0.0 s) and extended up to 0.1 seconds less than the mean duration of all annotations present in the recording (tmax=mean_duration - 0.1 s). A baseline correction was applied using a post-stimulus time window from 0.1 to 0.5 seconds after the event onset, ensuring that the baseline was not affected by pre-stimulus noise and occurred before the expected onset of motor imagery-related activity.

C. Feature Extraction

The preprocessed EEG epochs for the three classes formed the input for five feature extraction methods. Each employed a distinct CSP variant and processed the epochs independently to generate a feature matrix. The resulting features capture characteristics optimized by the respective CSP approach, log-variances or kernel projections [13].

1) Standard Common Spatial Pattern

Standard Common Spatial Pattern (CSP) employs spatial filters that transform multi-channel EEG signals into components. These filters are designed such that the variance of the projected components is maximized for one class while minimized for others [20].

In this study, we employed a multi-class CSP approach using a one-vs-rest (OvR) strategy. For each class, binary labels distinguished it from the others. The CSP algorithm was fitted for each binary problem using the corresponding epochs and labels. Spatial filters demonstrating the greatest class-discriminative variance were selected from the learned filter matrix. These filters were applied to each epoch, and features were computed as the log-variance of the projected components. Features from each OvR problem were concatenated to form the final feature matrix for the classifier.

2) Regularized Common Spatial Pattern

Regularized Common Spatial Pattern (RCSP) addresses standard CSP's sensitivity to noise and overfitting by introducing regularization during computation, typically by modifying covariance matrices [10]. We utilized the CSP implementation with automatic regularization via the Ledoit-Wolf shrinkage method to stabilize covariance estimation. The multi-class OvR strategy, selection of filters, feature computation, and feature concatenation followed the same procedure described for the Standard CSP.

3) Filter Bank Common Spatial Pattern

Filter Bank Common Spatial Pattern (FBCSP) aims to use discriminative information across multiple frequency bands. It first decomposes the EEG signal into several sub-bands using bandpass filters, then applies a CSP variant independently to each band, and finally combines the extracted features [13].

In this study, preprocessed EEG epochs were filtered into five bands (8–12 Hz, 12–16 Hz, 16–20 Hz, 20–24 Hz, 24–30 Hz) using a Butterworth filter. For each band, an OvR RCSP procedure (as described for RCSP) was applied independently. This involved selecting the most class-discriminative spatial filters for each OvR problem within the band, computing log-variance features, and concatenating these features across classes for that band. Finally, the feature sets from all five bands were merged to form the input feature matrix.

4) Sparse Common Spatial Pattern

Sparse Common Spatial Pattern (Sparse) addresses the dense filters produced by standard CSP by adding a sparsity-inducing penalty during optimization. This encourages many filter weights to be zero, performing implicit channel selection and highlighting critical electrode locations [15].

We implemented a multi-class Sparse CSP using an OvR strategy combined with sequential filter extraction via deflation. For each OvR problem, average data representations per class were used to compute trace-normalized covariance matrices. Spatial filters were found by minimizing an objective function representing the negative variance difference between classes, penalized by a smoothed L1-norm to promote sparsity, under a variance normalization constraint. This optimization used the SLSQP algorithm, initialized with a standard CSP solution. To extract multiple filters per OvR problem, a deflation technique sequentially updated the covariance matrices after each filter was found. Features were computed as the log-variance of the original epoch data projected onto these sparse filters, following the method described for the Standard CSP. The final feature matrix, containing features from all OvR problems, was standardized using Z-score normalization.

5) Kernel Common Spatial Pattern

Kernel Common Spatial Pattern (Kernel) extends CSP using kernel methods to capture non-linear relationships. It typically operates on epoch representations (like covariance matrices) rather than raw signals, using a kernel function to measure similarities and implicitly mapping data to a higher-dimensional Reproducing Kernel Hilbert Space (RKHS). Projections enhancing class separability are found within this space [14].

Each epoch was represented by its covariance matrix, estimated using Oracle Approximating Shrinkage (OAS) and then trace-normalized. For its flexibility, an RBF kernel was used to compute pairwise similarities between these covariance matrices based on the squared Frobenius norm distance, yielding a kernel matrix (with kernel width parameter set to 100.0). Feature extraction proceeded using an OvR strategy for each class: an eigenvalue decomposition was performed on the sub-kernel matrix corresponding only to the trials of that class. The eigenvectors corresponding to the largest eigenvalues (representing the principal directions in the kernel space) were retained. Features for an epoch were obtained by projecting its kernel vector (representing similarities to all epochs of a target class) onto that class's principal directions. These projection features were concatenated across all classes for each epoch, forming the final feature vector.

D. Classification

All experiments were conducted on a system equipped with an AMD Ryzen 5 8645HS processor, 16 GB of RAM, and an integrated AMD Radeon 760M GPU. We systematically evaluated each CSP variant by extracting features using 1, 3, 5, ..., up to 17 components. For each resulting feature set, data was partitioned into training (80%) and testing (20%) using a fixed split [21]. Classification involved standardizing these features, followed by training a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel (C=10, class_weight= 'balanced') exclusively on the training data. The performance of the trained model was then evaluated on the held-out test set using accuracy, F1-score, ROC AUC, Kappa, and confusion matrices to assess generalization and the impact of component number per variant.

III. RESULTS AND DISCUSSION

The analysis examines three key aspects: classification performance, computational cost, and the performance-cost trade-off.

A. Classification Performance

The evaluation was performed in a subject-dependent manner. As summarized in Table 1, CSP and RCSP achieved the highest peak average test accuracies (~97.8% at 15 components), leading across metrics with F1 scores ~0.957, Kappa scores ~0.947, and near-perfect ROC AUC (~0.998). Their performance was nearly identical, although RCSP held a marginal accuracy advantage specifically at 11 components (0.964 vs 0.961). FBCSP also demonstrated strong results, peaking earlier at 11 components with ~96.7% accuracy, ~0.946 F1, ~0.933 Kappa, and ~0.995 ROC AUC. Sparse reached its peak accuracy (~95.3%) at 15 components, notably achieving a high Kappa (~0.938) relative to its accuracy/F1 (~0.950), and a strong ROC AUC (~0.995). Conversely, Kernel's best performance (~92.8% Acc, ~0.942 F1, ~0.919 Kappa, ~0.994 ROC AUC) occurred very early at 3 components, indicating optimal feature representation at low complexity despite lower peak values compared to others.

Analysis of trends in Fig. 1 showed F1-scores closely mirroring accuracy for all methods. CSP and RCSP performance rose sharply, plateaued between 3-9 components, then increased slowly to their peak at 15. FBCSP improved gradually, peaking at 11 components. Sparse started with the lowest performance but showed steady improvement, surpassing Kernel around 7-9 components, though it remained consistently below the performance levels of the others. Kernel peaked early at 3 components and fluctuated lower afterwards.

TABLE I. SUMMARY OF PEAK AVERAGE TEST PERFORMANCE METRICS ACHIEVED BY EACH METHOD

Method	*N-Comp*	*Features*	*Accuracy*	*F1-Score*	*ROC AUC*	*Kappa*
CSP	15	45	0.9777	0.9566	0.9979	0.9467
RCSP	15	45	0.9777	0.9566	0.9979	0.9467
FBCSP	11	165	0.9666	0.9462	0.9952	0.9333
SPARSE	15	45	0.9527	0.9498	0.9949	0.9377
KERNEL	3	9	0.9277	0.9416	0.9944	0.9192

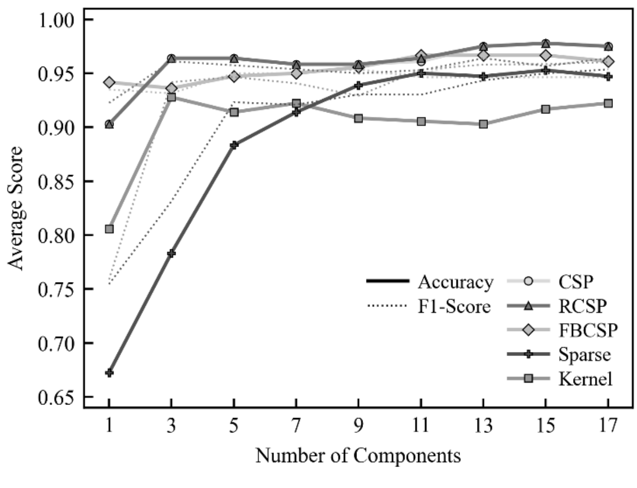

Figure 1. Performance trends for five spatial filtering methods. Average test accuracy (solid lines with markers) and average F1-score (thinner, dotted lines without markers are plotted together against the number of spatial filter components. Each marker represents a different method, as detailed in the legend.

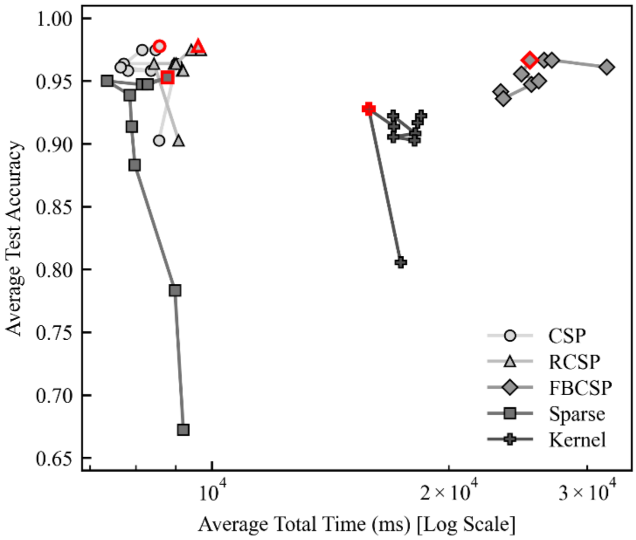

Figure 2. Average test accuracy vs. average total computation time (log scale). Points represent different numbers of components for each method (distinguished by marker); the marker outlined in red highlights the peak average test accuracy achieved by that method. The lines connect points for the same method.

B. Computational Cost

Computational time analysis in Fig. 3 reveals significant differences. FBCSP and Kernel are the slowest, bottlenecked by extensive feature extraction due to multi-filterbanks and kernel computations, respectively. Kernel's feature extraction time is relatively constant with varying numbers of components, whereas FBCSP's feature extraction and SVM training times increase. In contrast, CSP, RCSP, and Sparse are much more efficient (<10s average total time). CSP and RCSP show balanced time components, with feature extraction being the largest but far less than Kernel CSP and FBCSP, exhibiting modest scaling. Sparse CSP has the lowest feature extraction time, but SVM training is its primary and increasing cost with the number of components. Preprocessing time is consistently minimal. This confirms feature extraction bottlenecks Kernel and FBCSP, SVM training bottlenecks Sparse, and highlights the superior efficiency of standard CSP and RCSP.

C. Performance vs. Cost Trade-off

Figure 2 highlights the accuracy-cost trade-off, favoring methods in the top-left (high accuracy, low cost). RCSP and CSP demonstrate the best balance, achieving top peak accuracies (~97.8% at 15 components) with significantly lower computation time (~9s) than other high performers. FBCSP, while competitive in accuracy (~96.7% at 11 components), incurs the highest time cost (~23-32s) due to intensive feature extraction. Kernel offers moderate accuracy (~92.8% at 3 components) at moderate cost (~16-18s), also dominated by feature extraction, making it less efficient than CSP and RCSP. Sparse is faster than FBCSP and Kernel, peaking at ~95.3% accuracy (15 components); specifically at 11 components, it's fastest (~7.4s, ~95.0% acc) but CSP yields higher accuracy (~96.1%) for minimal extra time (~7.6s). Notably, CSP and RCSP performance plateaus between 3-9 components, offering >96% accuracy in <9s, suggesting a practical near-optimal point for real-time use, as further accuracy gains come at a substantial time cost.

D. Limitations

These findings are based on data averaged across 10 randomly selected subjects using a particular MI paradigm and dataset. Generalizability to other datasets, paradigms, or larger populations requires further investigation. The SVM classifier used fixed hyperparameters and performance might vary with different classifiers. Furthermore, the range of components was limited by the number of channels, and exploring different parameterizations within methods (e.g., regularization parameters for RCSP/SPARSE, kernel types/parameters for Kernel CSP, filter bank design for FBCSP) was beyond the scope of this work.

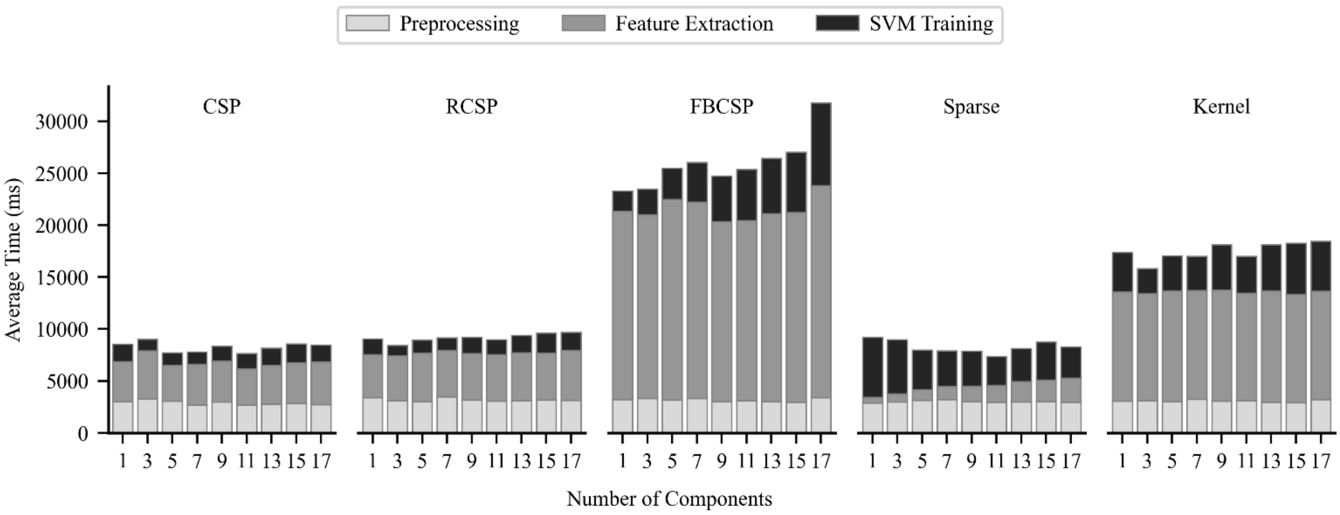

Figure 3. Breakdown of computation time into Preprocessing, Feature Extraction, and SVM Training phases across different numbers of components for each method.

IV. CONCLUSION

This study investigated the performance and computational cost of five CSP variants including standard CSP, RCSP, FBCSP, Sparse, and Kernel for multi-class MI-BCI using PhysioNet data and a standardized SVM evaluation across 10 subjects. Standard CSP and RCSP achieved the highest accuracy (~97.8%) with low computational demands. FBCSP was also accurate (~96.7%) but computationally expensive. Sparse CSP was faster but less accurate (~95.3%), while Kernel CSP had the lowest accuracy (~92.8%) and moderate cost. Considering the accuracy-efficiency trade-off, standard CSP and RCSP proved most promising, delivering >96% accuracy quickly with 5-9 components, suggesting suitability for real-time systems. While limited by the specific dataset and methods used, this work provides valuable benchmarks, highlighting standard CSP and RCSP as effective and efficient choices for multi-class MI under these conditions.

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